



National University

of Computer & Emerging Sciences Peshawar Campus



Name: Arsom Ali

Roll No: 23-P0038

Program: BCS/BSE/AI
Semester: Fall – 2023
Time Allowed: 1 hour
Course: NS1001 Applied Physic

Examination: Midterm II
Total Marks: 15 Weightage: 15
Date: Nov 14, 2023
Instructor: Engr M Asif Khan

NOTE: Attempt all questions.

Question 1

[5M]

A man wiping water from a floor with a wiper causes it to undergo two displacements. The first has a magnitude of 150cm and makes an angle of 120° with the positive x-axis. The resultant displacement has a magnitude of 140cm and is directed at an angle of 35° to the positive x-axis. Calculate the magnitude and direction of the second displacement.

Question 2

Design equations for the position, velocity, and acceleration of a particle performing simple harmonic motion

[5M]

Question 3

[5M]

A rock is thrown upward from the level ground in a way that the maximum height of its flight is equal to its horizontal range. Calculate the launching angle of the rock. Moreover comment on your result (whether the same or different) if the rock was thrown in another planet.

The End